



Article

Auditing Data Leakage and Temporal Misalignment in Machine-Learning Prognosis of Breast Cancer: A Reproducible Censoring-Aware Reanalysis

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Abstract

Background: Near-perfect machine-learning performance on public clinical datasets can arise from data leakage or predictors unavailable at the intended prediction time. We audited a breast cancer vital-status workflow and established a censoring-aware benchmark. **Methods:** The public SEER-derived dataset contained 4024 women diagnosed in 2006–2010. After removing one duplicate, 616 of 4023 records had status “Dead”. We reproduced a support-vector machine workflow that oversampled the complete dataset before an 80:20 split, quantified exact train-test row overlap, and repeated the analysis with training-only oversampling. A baseline-only support-vector machine and a penalized Weibull accelerated failure-time model were then evaluated using training-only tuning and an untouched test set. **Results:** Pre-split oversampling placed duplicate feature rows from 50.5% of test observations in training and yielded an area under the receiver operating characteristic curve (AUROC) of 0.998. Splitting first eliminated overlap and reduced AUROC to 0.715 under the retained legacy model. The tuned baseline-only model achieved AUROC 0.745 (95% confidence interval 0.696–0.793). The survival model achieved a test concordance index of 0.753 (0.708–0.797). **Conclusions:** Resampling order and post-baseline information materially inflated apparent performance. The corrected results support moderate internal prognostic discrimination, not breast cancer diagnosis, triple-negative subtype classification, or clinical deployment.



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Keywords: breast cancer; data leakage; oversampling; survival analysis; machine learning; reproducibility; censoring; clinical prediction model

1. Introduction

Machine learning is increasingly used to derive diagnostic and prognostic models from clinical records, imaging, molecular profiles, and population registries [1–3]. Its appeal is clear: flexible algorithms can represent nonlinear associations, interactions, and

high-dimensional data structures that may be difficult to specify in conventional regression models. Nevertheless, methodological flexibility also creates many opportunities for biased evaluation. A high reported accuracy is meaningful only when the prediction target, prediction time, cohort, preprocessing, validation design, and metric all correspond to the intended clinical use.

Data leakage is one of the most damaging and frequently under-reported threats to machine-learning validity. Leakage occurs when model development uses information that would not be available when predictions are made in practice or when observations in the evaluation set influence training [4,5]. The resulting model may appear highly accurate while primarily exploiting duplicated observations, post-outcome information, preprocessing fitted on the complete dataset, or other shortcuts. Leakage has been documented across scientific fields and can produce dramatic failures of reproducibility [5].

Class imbalance is a common setting in which leakage can be introduced. Clinical outcomes such as death, recurrence, or rare disease may comprise a minority of observations. Random oversampling duplicates minority-class records to balance the training data. The method is legitimate when applied only to the training partition or within training folds. If it is applied before the train-test split, however, copies of the same observation may be assigned to both partitions. The model is then evaluated partly on records that it has already seen. Similar concerns apply to synthetic oversampling: related synthetic samples can contaminate validation if resampling precedes partitioning [6–8].

A second problem arises when the outcome is temporally misframed. A binary label describing whether a patient was alive or dead at the end of observed follow-up is not equivalent to a fixed-horizon risk outcome. Different patients have different observation durations, and patients recorded as alive are right-censored. Treating this endpoint as ordinary binary classification discards follow-up time and assumes that all labels have the same observation basis. The problem becomes more severe when survival duration itself is included among the predictors: a variable measured after baseline is then used to predict a status determined during that same interval.

Time-to-event methods explicitly model event time and censoring. Their evaluation requires measures designed for censored outcomes, such as a concordance index, time-specific Brier score, and calibration assessment [9–12]. Discrimination alone is insufficient. A model may rank patients well while systematically overestimating or underestimating their absolute risks, which can make it unsuitable for decision support [13,14].

Transparent reporting standards now emphasize these distinctions. TRIPOD+AI requires clear definitions of the target population, intended prediction time, outcome, candidate predictors, preprocessing, model-development procedures, validation type, discrimination, calibration, and uncertainty [15]. PROBAST provides a complementary framework for assessing risk of bias in prediction-model studies [16]. These standards are especially relevant when a computational workflow combines several endpoints or datasets under one clinical label.

The dataset examined in this study is the open-access “SEER Breast Cancer Data” collection deposited in IEEE DataPort [17]. It contains records for women already diagnosed with infiltrating duct and lobular carcinoma. It therefore cannot evaluate breast cancer detection because no cancer-free control group is present. It includes estrogen receptor (ER) and progesterone receptor (PR) status but no human epidermal growth factor receptor 2 (HER2) variable; triple-negative breast cancer (TNBC), which requires ER-negative, PR-negative, and HER2-negative status, cannot be identified.

The objective of this study was not to propose another high-performing classifier. Instead, we aimed to provide a reproducible biomedical-informatics audit of how

workflow order and temporal predictor availability change apparent performance. Our contributions were:

1. reproduction of a leakage-prone oversampling-before-splitting workflow;
2. direct quantification of exact feature-row overlap between training and test data;
3. a matched split-first analysis retaining the legacy encoding, predictors, and support-vector machine (SVM) settings;
4. a training-only tuned baseline SVM that excluded follow-up duration;
5. a deliberately non-deployable sensitivity analysis that quantified the apparent gain from adding follow-up duration;
6. a censoring-aware penalized Weibull accelerated failure-time (AFT) model with internal discrimination, prediction-error, calibration, and risk-stratification analyses; and
7. release of an executable analysis script, pinned environment, machine-readable results, and figure source data.

2. Materials and Methods

2.1. Study Design

We performed a reproducibility audit and secondary analysis of a publicly accessible, de-identified dataset. The study was organized around two related questions. First, how much did pre-split oversampling contaminate the evaluation and inflate apparent classification performance? Second, what performance remained when the task was restricted to baseline predictors and evaluated with a censoring-aware survival model?

Figure 1 summarizes the analysis. The leakage audit reproduced the documented legacy sequence and then altered one component at a time. The primary prognostic analysis used an independent 80:20 development-test partition and a parametric AFT model. All model selection, preprocessing, and tuning operations were restricted to development data.

2.2. Data Source and Provenance

The “SEER Breast Cancer Data” dataset was deposited by Teng in IEEE DataPort in 2019 (DOI: 10.21227/a9qy-ph35) [17]. According to its metadata, the dataset was extracted from the November 2017 update of the US National Cancer Institute Surveillance, Epidemiology, and End Results program. It includes female patients diagnosed during 2006–2010 with infiltrating duct and lobular carcinoma (histology code 8522/3). The dataset creator excluded records with unknown tumor size, unknown numbers of examined or positive regional lymph nodes, or survival shorter than one month.

The downloaded comma-separated file contained 4024 rows and 16 columns. The variables were age, race, marital status, T stage, N stage, sixth-edition cancer stage, differentiation, grade, summary stage, tumor size, ER status, PR status, regional nodes examined, regional nodes positive, survival months, and vital status. The complete variable dictionary is provided in Appendix A Table A1.

The source file did not contain a patient identifier, institution identifier, treatment variables, comorbidity variables, cause of death, HER2 status, or timestamps beyond survival duration. Consequently, patient-level linkage to another dataset, TNBC identification, breast-cancer-specific mortality modeling, and external institutional validation were not possible.

2.3. Data Integrity and Cohort Construction

Column names and string values were stripped of surrounding whitespace. Missingness was assessed before modeling. Exact duplicate rows were counted using all distributed variables and removed before analysis. One duplicate was identified, leaving 4023 unique rows. No missing values were present in the distributed file.

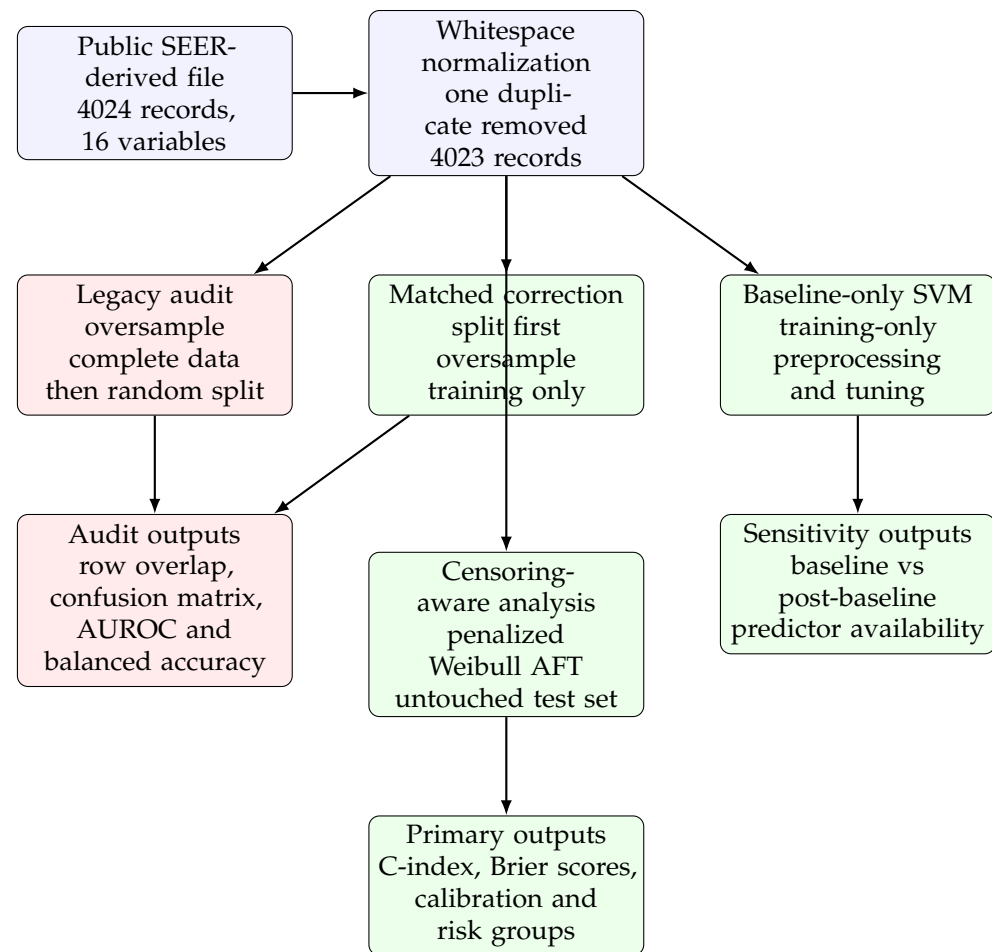


Figure 1. Analysis framework. Red denotes the intentionally reproduced leakage-prone workflow; green denotes leakage-controlled or censoring-aware analyses. The four classification workflows were used to separate record contamination from temporal predictor leakage. The survival model was the primary prognostic analysis.

Near-duplicate checks were also performed to avoid confusing ordinary registry ties with duplicate patients. After removal of the one full-row duplicate, 13 records shared all baseline variables with another record, with each tied group containing two records. These records differed in survival time or status and were retained because the public file contains no patient identifier and identical baseline covariates can occur among distinct patients. No pairs were duplicated after ignoring status alone. The audit therefore removed only the single exact full-row duplicate.

The event indicator was defined as one when Status equaled “Dead” and zero when it equaled “Alive”. Survival time was the distributed `Survival Months` variable. A record with status “Alive” was treated as right-censored at its recorded survival month. We did not reinterpret the status label as breast-cancer-specific death because cause of death was not available.

2.4. Definition of the Prediction Problems

2.4.1. Legacy Vital-Status Classification

The legacy task predicted end-of-recorded-follow-up vital status as a binary endpoint. This analysis was reproduced only to diagnose methodological failure. It was not considered a valid diagnostic task or the preferred prognostic formulation. The predictor matrix included survival months, which is unavailable at the time of diagnosis.

2.4.2. Baseline-Only Vital-Status Classification

The corrected classification sensitivity analysis excluded status, the derived event indicator, the derived time variable, and survival months from the predictors. It asked whether baseline demographic, stage, tumor, nodal, and receptor variables could distinguish records ultimately labeled dead from records labeled alive. Because follow-up duration remained heterogeneous, this analysis was considered secondary to the survival model.

2.4.3. Censoring-Aware Prognosis

The primary prognostic estimand was relative time to recorded all-cause death from diagnosis, conditional on the available baseline variables. Predictions were evaluated at 36, 60, and 84 months. The study did not define a treatment decision threshold or claim clinical utility.

2.5. Reproduction of the Leakage-Prone Workflow

The legacy notebook used manually assigned integer codes for categorical variables. Race, marital status, T stage, N stage, sixth-edition stage, differentiation, grade, summary stage, ER status, and PR status were mapped to integers. This coding imposes artificial numerical spacing on nominal variables, but it was retained for faithful reproduction.

Random oversampling was applied to the complete feature matrix and target using random seed 0. The resampled data were then divided into 80% training and 20% test partitions using `random_state=0` without stratification. An SVM with radial basis function kernel, $C = 100$, and $\gamma = 1$ was fitted. The same feature set, including survival months, was retained.

Random oversampling rather than SMOTE was used in the corrected legacy analysis because the purpose was to isolate preprocessing order while holding the reproduced resampling method constant. The baseline-only tuned SVM did not use oversampling; it used class weights within a training-only preprocessing pipeline.

2.6. Quantification of Exact Train-Test Overlap

Each feature row was hashed after partitioning. Let $\mathcal{H}_{train}$ be the set of hashes for training feature rows and h_i be the hash of test row i . The exact overlap fraction was

$$\text{Overlap} = \frac{1}{n_{test}} \sum_{i=1}^{n_{test}} I(h_i \in \mathcal{H}_{train}), \quad (1)$$

where $I(\cdot)$ is the indicator function. Because the single duplicate in the original file was removed before modeling, overlap in the oversampled analysis primarily represents copies introduced by resampling before partitioning.

2.7. Matched Split-First Legacy Analysis

To isolate the practical effect of moving oversampling behind the train-test boundary, we repeated the legacy analysis with one change: the original unique records were split first, and random oversampling was applied only to the training partition. Integer encoding, survival-month inclusion, SVM hyperparameters, test fraction, and random seed were retained. The resulting test set was not oversampled.

This comparison is intentionally diagnostic rather than a formal paired experiment. Pre-split oversampling changes the size and composition of the test set; therefore, the two AUROCs do not arise from identical observations. The direct row-overlap measure is the principal evidence of contamination, while the performance difference illustrates its practical consequence.

2.8. Leakage-Controlled Baseline SVM

The baseline-only model used a scikit-learn pipeline [18]. Numeric predictors were standardized using means and standard deviations estimated from training data. Categorical predictors were one-hot encoded with the first level omitted and unknown levels ignored. The SVM used class weights inversely proportional to class frequencies.

The dataset was split into an 80% training partition and a 20% test partition with `random_state=2026`, stratified by the event indicator. Hyperparameters were selected within the training partition by five-fold stratified cross-validation:

$$C \in \{0.1, 1, 10\}, \quad \gamma \in \{\text{scale}, 0.01, 0.1\}. \quad (2)$$

Mean cross-validated AUROC was the selection criterion. The preprocessing pipeline and selected SVM were refitted on the complete training partition and evaluated once on the untouched test partition.

2.9. Post-Baseline Predictor Sensitivity Analysis

To quantify temporal predictor leakage separately from row duplication, the corrected SVM pipeline was repeated with survival months added to the predictor set. Partitioning, preprocessing, cross-validation, tuning grid, and evaluation were otherwise identical. This model is explicitly non-deployable at diagnosis because its additional predictor is observed only during follow-up.

The role of `Survival Months` differs between the classification and survival analyses. In classification, it is a post-baseline input feature and therefore a leakage variable for diagnosis-time prediction. In the survival analysis below, the same column is not a predictor; it is the duration outcome T_i paired with the event/censoring indicator.

2.10. Classification Performance Measures

We reported the confusion matrix, accuracy, balanced accuracy, sensitivity, specificity, F1 score, and AUROC. For death as the positive class,

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}, \quad (3)$$

and

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}. \quad (4)$$

Balanced accuracy was included because ordinary accuracy can remain high when the classifier predicts only the majority class. AUROC and balanced-accuracy 95% confidence intervals were estimated from 1000 nonparametric bootstrap resamples of the test observations. Resamples containing only one outcome class were omitted for AUROC. Sensitivity and specificity 95% confidence intervals were estimated using Wilson binomial intervals and are reported in Appendix E Table A5.

Formal DeLong tests were not used for the AUROC differences. The leakage-prone and split-first legacy workflows did not evaluate identical independent test observations because pre-split oversampling changed the test-set composition and introduced direct dependence between training and test records. The baseline-only and follow-up-augmented analyses used the same test split, but the added follow-up variable is temporally invalid for diagnosis-time prediction; a statistical superiority test would not answer a clinically meaningful model-comparison question.

2.11. Penalized Weibull Accelerated Failure-Time Model

The primary survival model was a penalized Weibull AFT model fitted with life-lines [19]. This model was chosen a priori as a transparent censoring-aware benchmark because its coefficients have a direct time-ratio interpretation and it provides absolute survival estimates at clinically interpretable horizons. The analysis was not designed as a broad survival-algorithm competition.

AFT models relate predictors to the logarithm of event time. For subject i ,

$$\log(T_i) = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \sigma \epsilon_i, \quad (5)$$

where T_i is survival time, $\mathbf{x}_i$ is the predictor vector, $\boldsymbol{\beta}$ contains model coefficients, and ϵ_i follows the error distribution implied by the Weibull model. Thus Survival Months legitimately defines T_i in the survival model while remaining excluded from $\mathbf{x}_i$. Exponentiated coefficients are acceleration factors. Values greater than one correspond to longer predicted survival time; values less than one correspond to shorter predicted survival time, conditional on the other variables. These are associations, not causal effects.

Candidate numeric predictors were age, tumor size, regional nodes examined, and regional nodes positive. Candidate categorical predictors were race, marital status, sixth-edition stage, grade, ER status, and PR status. Redundant stage representations (T stage, N stage, summary stage) and the duplicate differentiation representation were excluded from the primary survival model to limit collinearity and avoid entering multiple encodings of the same clinical construct.

Proportional-hazards assumptions were not a prerequisite for this model because the selected estimand and model form were AFT rather than Cox proportional hazards. If a Cox model were used as a primary comparator, proportional-hazards diagnostics would be required.

Numeric predictors were standardized using training-set parameters. Categorical predictors were dummy encoded using training-defined levels. The data were split into 3218 training and 805 test observations, stratified by event status, using random seed 2026.

2.12. Survival-Model Tuning

The penalization strength and L1 ratio were selected using five-fold cross-validation in the training partition. Candidate combinations were

$$\text{penalizer} \in \{0.001, 0.01, 0.1, 1.0\}, \quad \text{L1 ratio} \in \{0, 0.5\}. \quad (6)$$

Each fold recomputed numeric scaling and categorical design matrices using only the fold-training observations. Mean Harrell concordance was the selection criterion. The selected model was then fitted to the complete training partition and applied to the test partition.

2.13. Survival-Model Evaluation

Discrimination was evaluated with Harrell's concordance index [9]. A 95% confidence interval was obtained from 1000 nonparametric bootstrap resamples of the test set.

Prediction error at horizons $t \in \{36, 60, 84\}$ months was evaluated using an inverse-probability-of-censoring weighted Brier score [10,11]. In general form,

$$\text{BS}(t) = \frac{1}{n} \sum_{i=1}^n w_i(t) \left[I(T_i > t) - \hat{S}_i(t) \right]^2, \quad (7)$$

where $\hat{S}_i(t)$ is predicted survival probability and $w_i(t)$ is estimated from the censoring distribution in the training data.

Calibration at 60 months was summarized by dividing test observations into quintiles of predicted mortality risk. For each quintile, mean predicted risk was compared with one minus the Kaplan-Meier survival estimate at 60 months. This grouped plot was used as an exploratory calibration display; it does not establish strong calibration [14].

For descriptive risk stratification, the median predicted risk in the training set was used as a fixed threshold. Test observations were classified as lower or higher predicted risk, Kaplan-Meier curves were estimated, and a log-rank test was reported. The test was not used for predictor selection or tuning.

2.14. Software, Reproducibility, and Quality Controls

The analysis used Python 3.12.13, NumPy 2.4.6, pandas 2.3.3, scikit-learn 1.9.0, imbalanced-learn 0.14.2, lifelines 0.30.3, and Matplotlib 3.11.0 [8,18,19]. Random seeds were fixed. The raw CSV SHA-256 checksum was:

```
88a78a30d26c16887c32e61dacf708bac605b586fc05a353fba0cca04c5525a5.
```

The source script writes machine-readable JSON and CSV results and regenerates every analysis figure. It also writes the primary train-test partition manifest so that the internal holdout split can be reconstructed exactly. The script accepts the downloaded source-file location as a command-line argument so that the supporting package does not depend on the authors' directory structure. Preprocessing is explicitly separated between training and evaluation data. The analysis was rerun from the distributed CSV in a clean virtual environment using the pinned requirements supplied with this manuscript.

OpenAI Codex (GPT-5) assisted with manuscript restructuring, LaTeX troubleshooting, reproducibility checks, and execution of the supplied analysis script. It did not generate the source data or determine participant inclusion. All code, outputs, interpretations, and manuscript text were reviewed by the authors, who retain responsibility for the work.

3. Results

3.1. Cohort Characteristics

The source file contained 4024 records. One exact duplicate was removed, leaving 4023 unique observations. There were 616 recorded deaths (15.3%). Median recorded follow-up was 73 months (interquartile range (IQR), 56–90). Mean age was 54.0 years (standard deviation, 9.0), and median tumor size was 25 mm (IQR, 16–38).

Most records were White (84.8%), married (65.7%), and classified as regional rather than distant disease (97.7%). ER-negative and PR-negative status were recorded in 6.7% and 17.4%, respectively. Only 242 observations (6.0%) were both ER-negative and PR-negative, but HER2 was absent, so these observations cannot be designated TNBC.

Recorded mortality increased across the distributed sixth-edition stages: 7.4% in IIA, 11.9% in IIB, 17.5% in IIIA, 29.9% in IIIB, and 38.3% in IIIC (Table 1). These are crude end-of-follow-up proportions and do not account for censoring or variable follow-up time.

3.2. Direct Evidence of Train-Test Contamination

The leakage-prone workflow produced 1363 test observations after the complete dataset had been balanced by random oversampling. Of these test observations, 50.5% had an exact feature-row duplicate in training. The contamination was concentrated in records copied by minority-class oversampling.

Table 1. Characteristics of the public SEER-derived cohort after removal of one exact duplicate.

Characteristic	Value	Characteristic	Value
Records, <i>n</i>	4023	Recorded deaths, <i>n</i> (%)	616 (15.3)
Age, mean (SD), years	54.0 (9.0)	Follow-up, median (IQR), months	73 (56–90)
Tumor size, median (IQR), mm	25 (16–38)	Regional nodes positive, median (IQR)	1 (1–4)
<i>Race, n (%)</i>			
Black	291 (7.2)	White	3412 (84.8)
Other	320 (8.0)		
<i>Receptor combinations, n (%)</i>			
ER negative/PR negative	242 (6.0)	ER negative/PR positive	27 (0.7)
ER positive/PR negative	456 (11.3)	ER positive/PR positive	3298 (82.0)
<i>Sixth-edition stage, n; recorded deaths, n (%)</i>			
IIA	1304; 96 (7.4)	IIB	1130; 135 (11.9)
IIIA	1050; 184 (17.5)	IIIB	67; 20 (29.9)
IIIC	472; 181 (38.3)		

SD, standard deviation; IQR, interquartile range; ER, estrogen receptor; PR, progesterone receptor. Percentages may not sum to exactly 100 because of rounding.

The contaminated test set yielded AUROC 0.998 (95% confidence interval (CI), 0.995–0.999), balanced accuracy 0.996 (95% CI, 0.992–0.999), and overall accuracy 99.6%. The confusion matrix contained 669 true negatives, no false positives, 6 false negatives, and 688 true positives.

When the original observations were split before oversampling, exact row overlap fell to zero. With the same integer encoding, inclusion of survival months, SVM hyperparameters, and random seed, AUROC fell to 0.715 (95% CI, 0.674–0.757). At the default decision threshold, the model predicted every test observation as alive: all 677 alive records were true negatives, while all 128 deaths were false negatives. Consequently, accuracy was 84.1%, but sensitivity and F1 score were both zero and balanced accuracy was 0.500.

Figure 2 juxtaposes apparent discrimination and direct overlap. The figure is an audit display rather than a claim of paired performance comparison because the pre-split oversampling workflow also altered test-set composition.

3.3. Baseline-Only and Follow-Up-Augmented Classification

The leakage-controlled baseline-only SVM selected $C = 1$ and $\gamma = 0.01$. Its mean five-fold training AUROC was 0.733. On the untouched test set, AUROC was 0.745 (95% CI, 0.696–0.793), balanced accuracy was 0.681 (95% CI, 0.636–0.726), sensitivity was 0.634 (Wilson 95% CI, 0.546–0.714), specificity was 0.727 (Wilson 95% CI, 0.693–0.759), and F1 score was 0.403. The model identified 78 of 123 deaths and produced 186 false positives.

When survival months were added to the otherwise identical corrected pipeline, mean training cross-validated AUROC increased to 0.857 and test AUROC increased to 0.895 (95% CI, 0.861–0.926). Sensitivity increased to 0.813 (Wilson 95% CI, 0.735–0.872) and specificity to 0.845 (Wilson 95% CI, 0.815–0.870). The gain quantifies the predictive information carried by follow-up duration when it is used as a predictor; it cannot be interpreted as performance available at diagnosis.

Table 2 presents all classification metrics. The results illustrate why accuracy alone was misleading. The split-first legacy model retained 84.1% accuracy despite missing every death because 84% of its test records were alive.

Table 3 summarises the corresponding discrimination metrics and uncertainty for the four workflows.

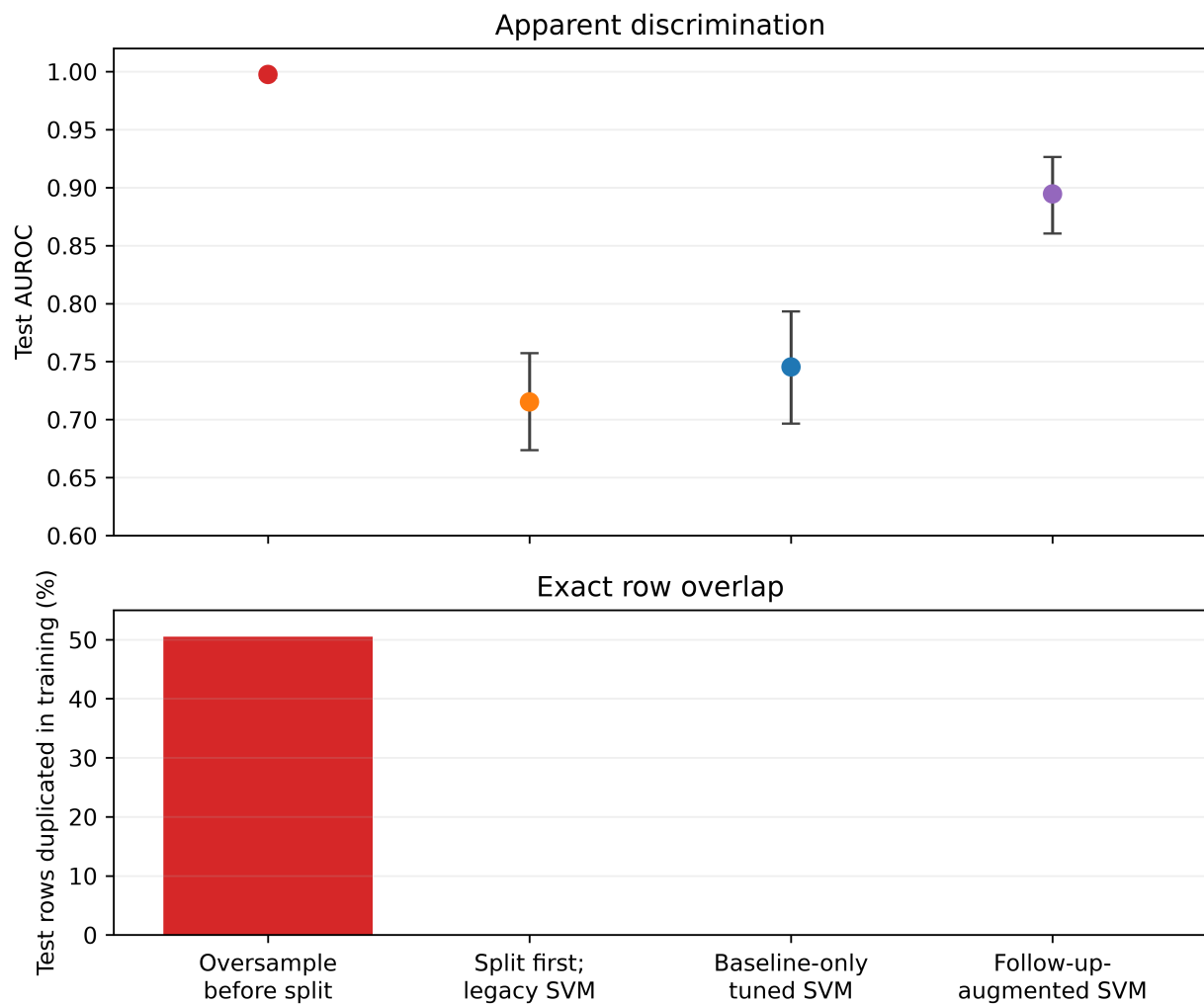


Figure 2. Summary of the classification audit. Points and bars in the (upper panel) show test AUROC and bootstrap 95% confidence intervals. The (lower panel) shows the percentage of test feature rows with an exact duplicate in training. The follow-up-augmented model is not deployable at diagnosis because survival duration is a post-baseline variable.

Table 2. Classification audit and sensitivity analyses.

Analysis	<i>n</i>	TN/FP/FN/TP	Acc.	Bal. Acc.	Sens.	Spec.	F1
Oversample before split; legacy predictors	1363	669/0/6/688	0.996	0.996	0.991	1.000	0.996
Split first; legacy predictors	805	677/0/128/0	0.841	0.500	0.000	1.000	0.000
Baseline-only tuned SVM	805	496/186/45/78	0.713	0.681	0.634	0.727	0.403
Follow-up-augmented tuned SVM	805	576/106/23/100	0.840	0.829	0.813	0.845	0.608

TN, true negative; FP, false positive; FN, false negative; TP, true positive; Acc., accuracy; Bal. Acc., balanced accuracy; Sens., sensitivity; Spec., specificity. Death was the positive class.

Table 3. Discrimination and uncertainty for the four classification workflows.

Analysis	Test AUROC (95% CI)	Balanced Accuracy (95% CI)	Row Overlap
Oversample before split; legacy predictors	0.998 (0.995–0.999)	0.996 (0.992–0.999)	50.5%
Split first; legacy predictors	0.715 (0.674–0.757)	0.500 (0.500–0.500)	0%
Baseline-only tuned SVM	0.745 (0.696–0.793)	0.681 (0.636–0.726)	0%
Follow-up-augmented tuned SVM	0.895 (0.861–0.926)	0.829 (0.792–0.863)	0%

AUROC, area under the receiver operating characteristic curve; CI, confidence interval.

3.4. Survival-Model Selection and Discrimination

The survival-model training and test sets contained 3218 and 805 observations, with 493 and 123 recorded deaths, respectively. Cross-validation selected penalizer 0.01 and L1 ratio 0, corresponding to ridge-type shrinkage. Mean cross-validated concordance was 0.727 (standard deviation 0.010). The complete tuning grid is reported in Appendix B Table A2.

On the untouched test set, the selected Weibull AFT model achieved a concordance index of 0.753 (95% CI, 0.708–0.797). This indicates moderate internal ranking of shorter versus longer observed survival times. It does not establish external transportability.

3.5. Time-Specific Prediction Error and Calibration

Test-set inverse-probability-of-censoring weighted Brier scores were 0.053 at 36 months, 0.097 at 60 months, and 0.116 at 84 months. The increase over time partly reflects the longer prediction horizon and increasing cumulative event incidence. No external benchmark model was available, so the Brier scores should not be interpreted as proof of clinical usefulness.

At 60 months, mean predicted mortality risks across quintiles were 0.033, 0.053, 0.077, 0.118, and 0.284. Kaplan-Meier observed risks were 0.021, 0.060, 0.070, 0.149, and 0.324, respectively. The model slightly overpredicted risk in the lowest quintile and underpredicted risk in the two highest quintiles (Figure 3); the corresponding numeric values are reported in Table A3. With only 123 events in the test set, a flexible calibration curve and stable formal calibration intercept or slope were not estimated.

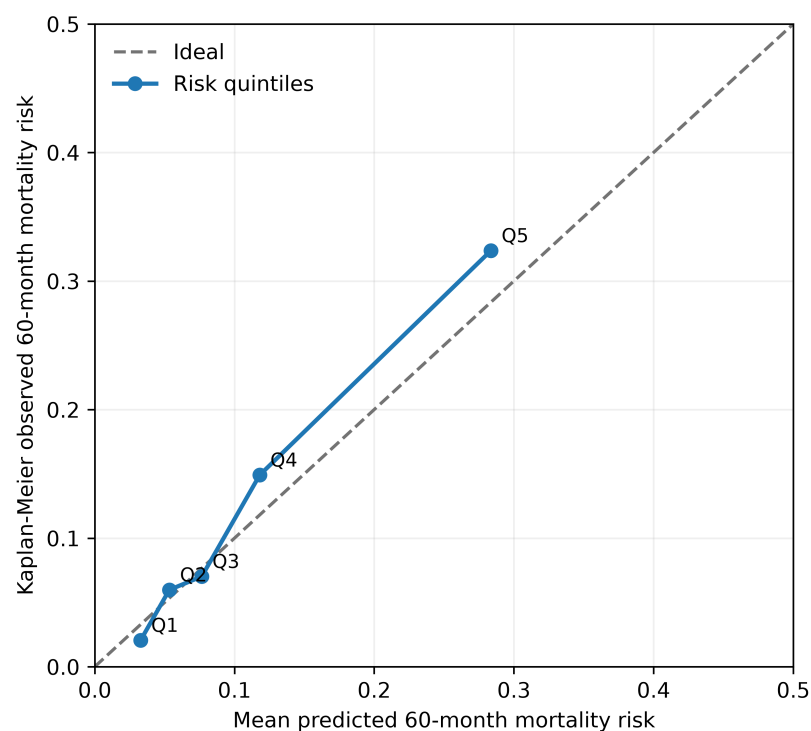


Figure 3. Internal test-set calibration at 60 months. Each point compares mean predicted mortality risk with Kaplan-Meier observed mortality risk in a quintile of predicted risk. The dashed line represents perfect agreement. Grouped calibration is descriptive and does not establish calibration in an external population.

3.6. Internal Risk Stratification

The median predicted risk in the training set divided the test set into 421 lower-risk and 384 higher-risk observations. Kaplan-Meier estimates separated throughout follow-up (descriptive log-rank $p = 8.34 \times 10^{-14}$; Figure 4). Because the model and threshold were derived from the same source population, this finding demonstrates internal risk stratification only.

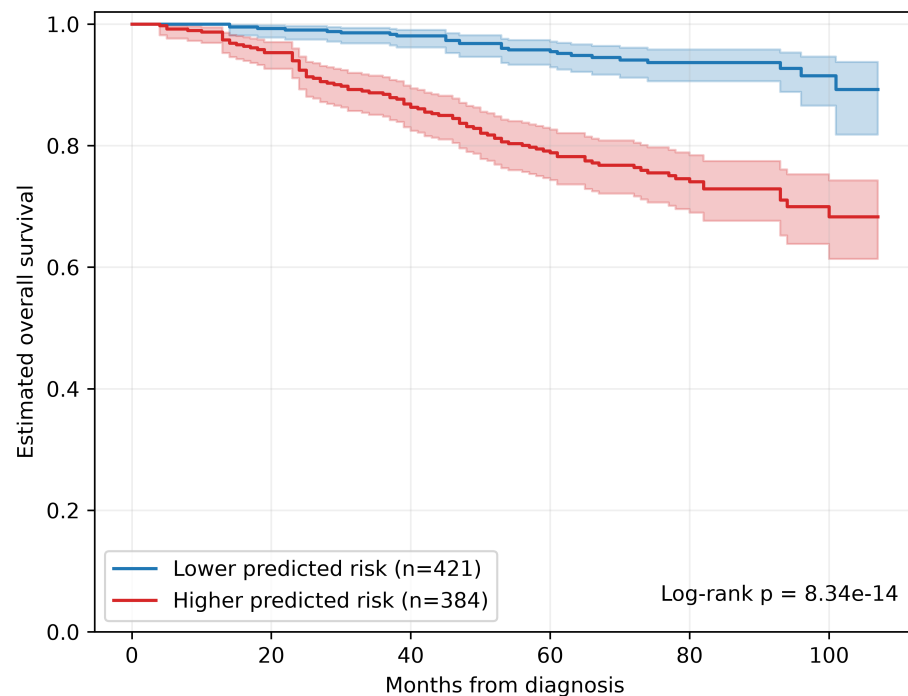


Figure 4. Kaplan-Meier overall-survival estimates in the internal test set, grouped using the median training-set predicted risk. Shaded regions are 95% confidence intervals. The log-rank test was descriptive and was not used for model development.

3.7. Model Coefficients

Figure 5 and Table A4 report the AFT acceleration factors, standardized log-time coefficients, and model-based 95% confidence intervals. Conditional on the other variables, later stages, higher grade, older age, and more positive nodes were associated with shorter predicted survival time. Positive ER and PR status were associated with longer predicted survival time. The tumor-size coefficient was close to null after the selected stage and nodal variables were included.

This tumor-size result should not be interpreted as evidence that tumor size is biologically or clinically unimportant. Tumor size is partly represented in anatomic stage and was correlated with nodal burden in the distributed cohort; median tumor size increased from 15 mm in stage IIA to 51 mm in stage IIIB, and the Spearman correlation between tumor size and positive-node count was 0.331. Conditional shrinkage, stage adjustment, and collinearity can therefore attenuate the separate tumor-size coefficient. Additional duplicate and collinearity diagnostics are summarized in Appendix E Table A6.

These coefficients should be interpreted cautiously. They are conditional associations from a selected public cohort, not causal effects. The dataset excludes records with missing nodal and tumor-size information and lacks treatment, comorbidity, and socioeconomic variables. Race and marital-status coefficients may reflect unmeasured structural and clinical factors and must not be interpreted biologically.

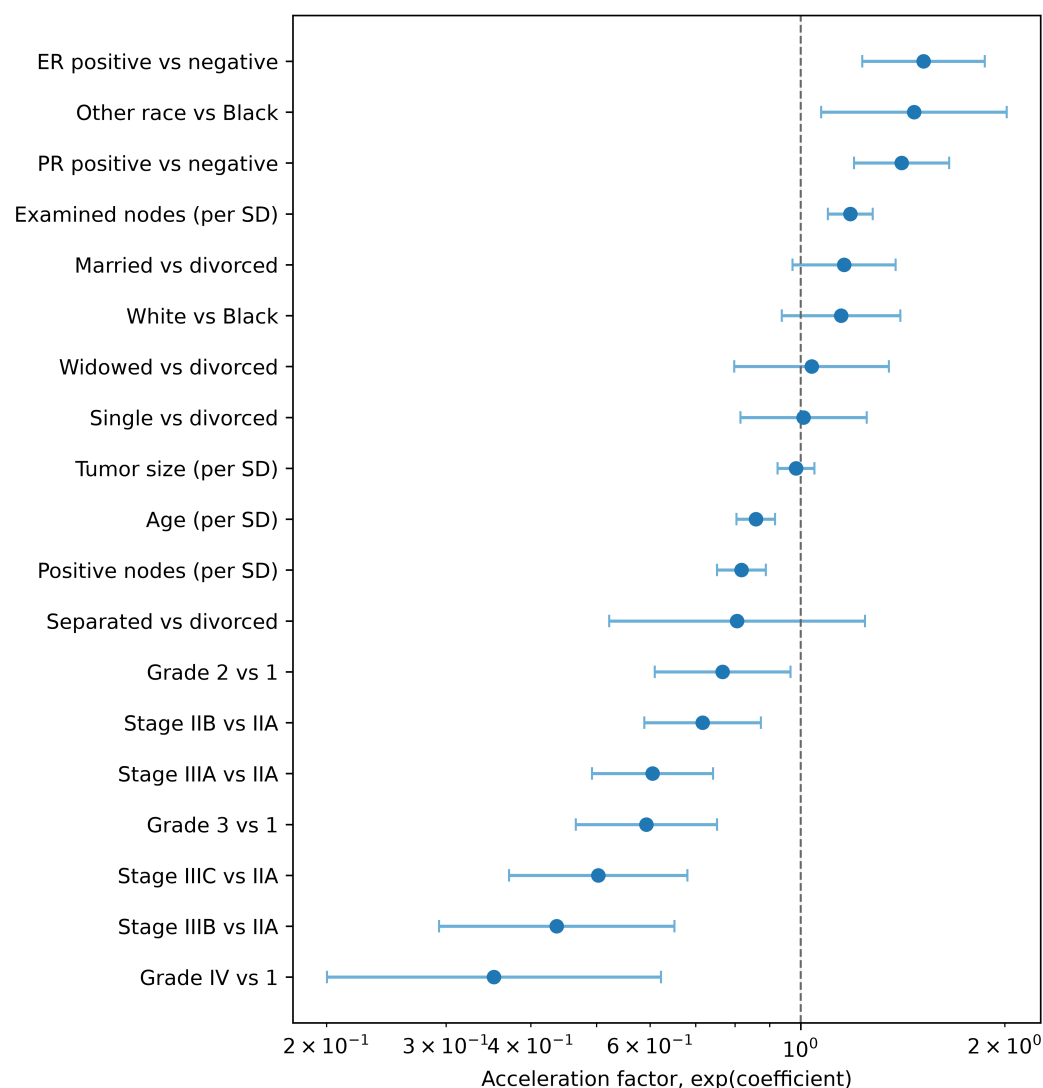


Figure 5. Acceleration factors from the penalized Weibull AFT model. Values below one indicate shorter predicted survival time and values above one indicate longer predicted survival time, conditional on other predictors. Numeric predictors are expressed per training-set standard deviation. Reference categories were stage IIA, grade 1, ER negative, PR negative, Black race, and divorced marital status. Intervals are model-based and do not include model-selection uncertainty.

4. Discussion

4.1. Principal Findings

This study found two distinct sources of performance inflation in a public breast cancer machine-learning workflow. First, random oversampling before partitioning created direct record contamination: exact copies of more than half of test observations were present in training. The contaminated evaluation produced an AUROC of 0.998. Moving oversampling behind the train-test boundary eliminated overlap and reduced AUROC to 0.715 under the otherwise retained legacy workflow. Second, including observed survival duration as a predictor materially improved vital-status classification, but that information is unavailable at diagnosis and therefore cannot support baseline prediction.

After both issues were addressed, the tuned baseline-only SVM achieved AUROC 0.745. A censoring-aware Weibull AFT model achieved test concordance 0.753 and provided time-specific risk estimates with moderate internal discrimination. Calibration analysis indicated some underprediction among higher-risk observations. The corrected results

are scientifically plausible and substantially more modest than the original near-perfect classification result.

4.2. Contribution to Biomedical Informatics

The main contribution is a reproducible audit pattern rather than a new algorithm. Biomedical-informatics studies often focus on model architecture while devoting less attention to the sequence of preprocessing, partitioning, resampling, feature definition, and evaluation. Yet these design choices determine whether a result estimates performance on unseen patients or merely reflects information recycled from model development.

The audit combines two complementary checks. Row hashing provides direct evidence of observation-level contamination. Temporal predictor review identifies variables that may be numerically predictive but unavailable at the intended prediction time. Either check alone would be incomplete: removing duplicated rows would not make follow-up duration a legitimate baseline predictor, and removing follow-up duration would not repair pre-split oversampling.

The proposed workflow is broadly applicable. Any study using oversampling, augmentation, multiple records per patient, repeated measurements, image derivatives, or laboratory replicates should verify that related observations remain within one partition. Any clinical model should also define the prediction timestamp and classify every predictor according to whether it is available at that time.

4.3. Why Pre-Split Oversampling Is Particularly Misleading

Random oversampling does not create new independent information; it repeats minority-class observations. When copies are generated before a random split, the probability that at least one copy crosses the train-test boundary increases with the number of repetitions. A flexible classifier can then memorize local neighborhoods around copied records. The nominal test set ceases to represent independent data.

The contaminated test set in this study was also artificially balanced, whereas the original event prevalence was 15.3%. This changes the interpretation of threshold-dependent metrics and accuracy. The near-perfect confusion matrix therefore reflected both dependence between partitions and a test distribution that no longer matched the source cohort.

The split-first legacy analysis provides a useful caution. Although the SVM produced no positive classifications, its accuracy remained 84.1%, approximately the proportion of alive records. A headline accuracy without the confusion matrix, sensitivity, or balanced accuracy would conceal complete failure to detect deaths. This is consistent with established concerns about evaluating imbalanced classifiers using accuracy alone [6,20].

4.4. Temporal Availability and Target Leakage

Predictor availability must be judged relative to a specified prediction time. Survival months is known only after follow-up has accrued. Using it to predict whether status is alive or dead at the end of that follow-up is analogous to using information from the outcome window to predict the outcome. The variable is not simply strongly associated with the label; it partly defines the opportunity for the event to be observed.

The corrected follow-up-augmented SVM achieved AUROC 0.895, compared with 0.745 for the baseline-only SVM. This difference quantifies how much apparent discrimination can be gained from temporally unavailable information even when the train-test split itself is clean. It should not be interpreted as a valid incremental-value analysis because the added variable cannot be measured at diagnosis. This distinction is central to clinical translation. A model intended for diagnosis, treatment planning, or prognosis at initial presentation may use only information available at that point. Variables collected after treatment, recurrence, or follow-up may be legitimate for dynamic prediction at later

landmarks, but then the target population, prediction time, and evaluation design must be redefined.

The same data column can therefore be invalid or valid depending on its role in the model. `Survival_Months` is leakage when entered as a feature for baseline classification, but it is the correct time-to-event outcome when paired with the censoring/event indicator in a survival analysis.

4.5. Why a Survival Model Is More Appropriate

The binary status label collapses event occurrence and follow-up duration. A patient alive after 12 months and a patient alive after 100 months receive the same label despite very different evidence regarding long-term survival. Similarly, a death at 10 months and a death at 90 months are treated identically. Time-to-event modeling retains this information and accounts for censoring.

The Weibull AFT model was selected because its coefficients have a direct time-ratio interpretation and because it can estimate survival probabilities at clinically interpretable horizons. Penalization reduced coefficient instability in the presence of correlated stage, nodal, and receptor information. This choice also avoided making proportional-hazards assumptions, which would have to be examined for a Cox proportional-hazards model but are not assumptions of an AFT model. Nevertheless, the Weibull distributional form is restrictive. We assessed its consequences only through internal discrimination, Brier scores, and grouped calibration; we did not perform a formal residual-based goodness-of-fit analysis. Future work should compare flexible parametric survival models, penalized Cox models with proportional-hazards diagnostics, random survival forests, and modern survival-learning methods under identical resampling and validation conditions.

The primary test concordance was 0.753; the baseline classification AUROC was 0.745. These metrics are not directly interchangeable. The concordance index compares risk ordering among usable pairs while respecting event times and censoring. The binary AUROC concerns end-of-recorded-follow-up status and ignores when events occurred.

4.6. Calibration and Clinical Interpretation

Discrimination is necessary but insufficient for clinical risk prediction. A model can rank patients correctly while giving inaccurate absolute risks. At 60 months, the present model underestimated observed mortality in the two highest predicted-risk quintiles. Such underprediction could matter if risk estimates were used to determine surveillance or treatment intensity.

The calibration plot is intentionally described as exploratory. With 123 events in the test set, detailed nonlinear calibration assessment would be unstable, and grouped points can hide within-group variation. Calibration intercept and slope were therefore not estimated as confirmatory statistics in the present internal test set. External validation should report calibration-in-the-large, calibration slope, a smoothed calibration curve, and uncertainty [14]. Decision-curve analysis should be considered only after a clinically meaningful decision and threshold range have been defined [21]; it was not included here because the audit did not define a treatment, surveillance, or triage decision.

4.7. Task Definition and the Limits of the Dataset

The source population already consisted of women diagnosed with breast cancer. Therefore, no result from this dataset can demonstrate detection of cancer among screened or symptomatic individuals. The outcome was recorded death, not malignant versus benign pathology. Diagnostic sensitivity and specificity are inappropriate descriptions of this task.

The dataset also cannot support TNBC-specific inference. TNBC requires absence of ER, PR, and HER2 expression. Although 242 records were ER-negative and PR-negative, HER2 was absent. Labeling those records as TNBC would introduce outcome misclassification. A TNBC study requires a cohort with verified receptor definitions, pathology procedures, treatment information, and a prespecified diagnostic or prognostic endpoint.

No external hospital cohort was available in the supplied project. Internal random holdout performance must not be described as external, multicenter, temporal, or prospective validation. Generalizability requires testing in a distinct population separated by institution, geography, or time, with a locked model and independently defined outcome.

4.8. Reproducibility and Reporting

The revised analysis is accompanied by the exact source-file checksum, executable script, pinned package versions, tuning output, coefficient table, classification metrics, calibration points, and machine-readable results. This level of documentation is intended to make both the error and its correction inspectable. The Supplementary Materials contain the complete analysis script, pinned environment, raw output data, and the primary partition manifest for full reproducibility.

The reporting structure follows the main principles of TRIPOD+AI [15]: explicit target definition, predictor timing, partitioning, model development, internal validation, performance uncertainty, and limitations. Appendix F provides a study-specific reproducibility checklist. A formal completed TRIPOD+AI checklist should accompany journal submission.

4.9. Strengths

Several features strengthen the analysis. The leakage mechanism was directly measured rather than inferred only from a drop in performance. The matched split-first analysis retained the legacy model specification, helping distinguish pipeline order from wholesale algorithm replacement. A separate baseline-only model quantified performance using temporally available predictors. The primary survival analysis retained censoring information and used training-only tuning. The raw data, software versions, random seeds, output tables, and figure-generation code were documented.

4.10. Limitations

This study also has substantial limitations. First, the dataset is a selected public extract rather than the complete SEER research database. Its preprocessing before deposition cannot be reconstructed from the CSV alone. Second, the status variable represents recorded all-cause death because cause of death was unavailable. Third, treatment, comorbidity, socioeconomic, genomic, HER2, and recurrence variables were absent. The model is therefore clinically incomplete.

Fourth, only internal random holdout evaluation was performed. Random splitting does not assess temporal drift, institutional heterogeneity, or geographic transportability. Nested cross-validation and repeated random splits were considered but were not used as the primary validation design because the study goal was to audit a specific workflow and provide a deterministic, inspectable internal benchmark. The exact seeds and primary

partition manifest are supplied, but sensitivity to alternative partitions remains a limitation. Fifth, the AFT analysis assumed a Weibull distribution. Although penalization was tuned by cross-validation, model-form uncertainty was not evaluated. Sixth, bootstrap intervals conditioned on the fitted model and test sample; they did not propagate uncertainty from predictor selection, preprocessing choices, or the initial train-test split. Bootstrap optimism correction was not performed because the main corrected analyses used an untouched internal test partition; however, optimism correction would be useful in a future development-only analysis.

Seventh, the grouped calibration display contained only 123 test events. It was insufficient for strong calibration claims. Eighth, the legacy analyses used arbitrary integer coding and fixed SVM hyperparameters because their purpose was reproduction, not optimal modeling. Ninth, subgroup performance was not estimated. Race categories in the dataset are coarse, and the analysis was not designed or powered as a fairness assessment across race, age, or stage strata. Tenth, model-explanation analyses such as SHAP or permutation importance were not performed for the baseline SVM because classifier interpretation was secondary to the leakage audit and because correlated clinical variables make single-feature importance unstable. The AFT coefficient table was used instead to provide an interpretable survival-model summary.

Finally, this study cannot determine whether the original oversampling order was the sole reason for every previously reported performance value. The supplied materials contained heterogeneous analyses and endpoints. Our claims are restricted to the workflows and data reproduced here.

4.11. Recommendations for Future Work

A publishable next-stage prognostic study should:

1. define a clinically meaningful prediction time and horizon before model development;
2. use a cohort with complete predictor provenance and clinically adjudicated outcomes;
3. justify sample size from the event prevalence, candidate parameters, and validation objective [22];
4. reserve a temporal or geographic external validation cohort;
5. use nested or repeated resampling when the goal is model selection rather than deterministic workflow audit;
6. keep all preprocessing, feature selection, resampling, and tuning inside development folds;
7. preserve patient or case groups during partitioning when repeated records are present;
8. compare a small number of prespecified survival models under the same validation design and check model-specific assumptions;
9. report discrimination, calibration, prediction error, and uncertainty;
10. assess clinical utility only after specifying a decision and plausible threshold range;
11. evaluate subgroup performance where sample sizes and metadata are sufficient; and
12. release a versioned code repository and immutable partition manifest.

For TNBC-specific work, verified ER, PR, and HER2 labels are a minimum requirement. For diagnostic imaging, a separate study should define the imaging modality, case-level split, reference standard, target classes, and appropriate classification, segmentation, or detection metrics. These tasks should not be combined with survival prediction unless a genuine patient-level multimodal design is available.

5. Conclusions

Pre-split oversampling caused direct train-test contamination in a public breast cancer vital-status workflow: exact copies of 50.5% of test feature rows appeared in training,

producing an apparent AUROC of 0.998. Splitting before oversampling eliminated overlap and sharply reduced performance under the retained legacy model. A second source of optimism arose from using survival duration, a post-baseline variable, to predict end-of-follow-up status.

After restricting predictors to baseline information and using training-only preprocessing and tuning, the SVM achieved moderate discrimination. A censoring-aware penalized Weibull model achieved test concordance 0.753 and showed some underprediction among higher-risk observations. These findings support a focused biomedical-informatics contribution on reproducibility, leakage, temporal alignment, and survival evaluation. They do not support breast cancer diagnosis, TNBC classification, external clinical validation, cost-effectiveness, or deployment.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/biomedinformatics6040060/s1>, the following supporting files accompany the submission: the complete Python reanalysis script, pinned Python environment, machine-readable result summary, primary train-test partition manifest, classification metrics, AFT tuning results, complete AFT coefficient output, 60-month calibration points, threshold-metric confidence intervals, and duplicate/collinearity checks. The exact file names are listed in the supplementary README.

Author Contributions: Conceptualization, M.A.S. and B.H.; methodology, M.A.S. and B.H.; software, M.A.S.; formal analysis, M.A.S.; data curation, M.A.S.; visualization, M.A.S.; writing—original draft preparation, M.A.S.; writing—review and editing, W.H., M.A., A.A. and B.H.; supervision, B.H.; project administration, B.H.; funding acquisition: L.T. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Not applicable to the computational analysis as reported: the study used a publicly accessible, de-identified secondary dataset and involved no participant recruitment, intervention, contact, or access to identifiable records.

Informed Consent Statement: Not applicable. No identifiable participant information is reported and no participants were contacted.

Data Availability Statement: The source dataset is publicly available from IEEE DataPort at <https://doi.org/10.21227/a9qy-ph35>. The exact CSV analyzed had SHA-256 checksum 88a78a30d26c16887c32e61dacf708bac605b586fc05a353fba0cca04c5525a5. The analysis script, package requirements, machine-readable outputs, figure source data, and primary partition manifest are supplied as supporting files. Before submission, the authors should archive the final package in a versioned public repository with a persistent identifier.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AFT	Accelerated failure time
AUROC	Area under the receiver operating characteristic curve
CI	Confidence interval
ER	Estrogen receptor
HER2	Human epidermal growth factor receptor 2

IPCW	Inverse probability of censoring weighting
IQR	Interquartile range
PR	Progesterone receptor
SEER	Surveillance, Epidemiology, and End Results
SVM	Support-vector machine
TNBC	Triple-negative breast cancer

Appendix A. Distributed Variable Dictionary

Table A1. Variables in the distributed public CSV and their role in the revised analysis.

Variable	Type	Use and Interpretation
Age	Numeric	Baseline predictor; years at diagnosis.
Race	Categorical	Baseline predictor with Black, White, and Other categories.
Marital Status	Categorical	Baseline predictor with divorced, married, separated, single, and widowed categories.
T Stage	Ordinal category	Distributed anatomic tumor stage; excluded from the primary AFT model because sixth-edition stage was used.
N Stage	Ordinal category	Distributed nodal stage; excluded from the primary AFT model because sixth-edition stage and nodal counts were used.
6th Stage	Ordinal category	Baseline predictor; stages IIA, IIB, IIIA, IIIB, and IIIC.
differentiate	Categorical	Duplicate representation of grade; excluded from the primary AFT model.
Grade	Ordinal category	Baseline predictor; grades 1–4.
A Stage	Categorical	Regional or distant summary stage; excluded from the primary AFT model to reduce redundant stage encoding.
Tumor Size	Numeric	Baseline predictor; millimeters.
Estrogen Status	Binary category	Baseline ER status.
Progesterone Status	Binary category	Baseline PR status.
Regional Node Examined	Numeric	Baseline predictor; number of regional nodes examined.
Reginol Node Positive	Numeric	Baseline predictor; number of positive regional nodes; spelling retained from source file.
Survival Months	Numeric	Excluded from baseline predictors because it is post-baseline leakage in classification; used legitimately as the event-time outcome in survival analysis.
Status	Binary category	Alive or Dead; converted to censoring/event indicator.

Appendix B. Complete Survival-Model Tuning Grid

Table A2. Five-fold training-only tuning results for the penalized Weibull AFT model.

Penalizer	L1 Ratio	Mean C-Index (SD)
0.01	0.0	0.7271 (0.0099)
0.001	0.0	0.7271 (0.0103)
0.001	0.5	0.7270 (0.0103)
0.01	0.5	0.7250 (0.0095)
0.1	0.0	0.7242 (0.0100)
1.0	0.0	0.7104 (0.0152)
0.1	0.5	0.6999 (0.0173)
1.0	0.5	0.5821 (0.1079)

AFT, accelerated failure time; SD, standard deviation across folds. The first row was selected.

Appendix C. Calibration Values

Table A3. Predicted and Kaplan-Meier observed 60-month mortality risk in test-set quintiles.

Risk Quintile	<i>n</i>	Mean Predicted Risk	Observed Risk
1	161	0.0328	0.0205
2	161	0.0534	0.0598
3	161	0.0766	0.0702
4	161	0.1181	0.1491
5	161	0.2836	0.3237

Appendix D. Complete AFT Coefficient Summary

Table A4. Penalized Weibull AFT standardized log-time coefficients and acceleration factors. Numeric predictors are expressed per training-set standard deviation; categorical predictors are contrasts against the stated reference category.

Predictor	Coefficient	Acceleration Factor	95% CI	<i>p</i> Value
Stage IIB vs. IIA	−0.333	0.717	0.588–0.874	0.0010
Stage IIIA vs. IIA	−0.503	0.605	0.493–0.743	<0.0001
Stage IIIB vs. IIA	−0.828	0.437	0.293–0.651	<0.0001
Stage IIIC vs. IIA	−0.687	0.503	0.372–0.681	<0.0001
Age, per SD	−0.152	0.859	0.804–0.917	<0.0001
ER positive vs. negative	0.417	1.517	1.232–1.868	<0.0001
Grade 2 vs. 1	−0.265	0.767	0.609–0.966	0.0242
Grade 3 vs. 1	−0.523	0.592	0.466–0.753	<0.0001
Grade IV vs. 1	−1.041	0.353	0.200–0.622	0.0003
Married vs. divorced	0.147	1.159	0.973–1.380	0.0983
Separated vs. divorced	−0.216	0.806	0.522–1.244	0.3297
Single vs. divorced	0.010	1.010	0.815–1.251	0.9279
Widowed vs. divorced	0.037	1.038	0.798–1.349	0.7811
PR positive vs. negative	0.343	1.409	1.199–1.656	<0.0001
Other race vs. Black	0.385	1.469	1.073–2.012	0.0166
White vs. Black	0.138	1.147	0.938–1.403	0.1801
Positive nodes, per SD	−0.201	0.818	0.753–0.888	<0.0001
Examined nodes, per SD	0.169	1.184	1.097–1.277	<0.0001
Tumor size, per SD	−0.016	0.984	0.925–1.048	0.6220

AFT, accelerated failure time; CI, confidence interval; SD, standard deviation; ER, estrogen receptor; PR, progesterone receptor. Model-based *p* values are descriptive and were not used for predictor selection.

Appendix E. Additional Sensitivity and Audit Checks

Table A5. Wilson 95% confidence intervals for threshold-dependent classification metrics.

Analysis	Sensitivity (95% CI)	Specificity (95% CI)
Oversample before split; legacy predictors	0.991 (0.981–0.996)	1.000 (0.994–1.000)
Split first; legacy predictors	0.000 (0.000–0.029)	1.000 (0.994–1.000)
Baseline-only tuned SVM	0.634 (0.546–0.714)	0.727 (0.693–0.759)
Follow-up-augmented tuned SVM	0.813 (0.735–0.872)	0.845 (0.815–0.870)

Table A6. Duplicate, near-duplicate, and collinearity checks relevant to reviewer-requested interpretation.

Check	Result and Interpretation
Exact full-row duplicates	One duplicate was identified across all distributed variables and removed before analysis.
Near-duplicates using baseline variables only	Thirteen records shared all baseline variables with another record; each tied group contained two records. These were retained because they differed in survival time or status and may represent distinct registry patients.
Duplicates after ignoring status only	None. No pair had identical covariates and survival time while differing only in vital status.
Tumor size vs. positive nodes	Spearman correlation 0.331; Pearson correlation 0.242.
Tumor size vs. examined nodes	Spearman correlation 0.128; Pearson correlation 0.104.
Positive vs. examined nodes	Spearman correlation 0.367; Pearson correlation 0.412.
Median tumor size by stage	15 mm (IIA), 30 mm (IIB), 36.5 mm (IIIA), 51 mm (IIIB), and 36.5 mm (IIIC).

Appendix F. Reproducibility Checklist

Table A7. Study-specific reproducibility and leakage-control checklist.

Item	Implementation
Data provenance	Dataset DOI, source description, inclusion criteria, and raw-file SHA-256 checksum reported.
Unit of analysis	One distributed row per patient record; one exact duplicate removed.
Prediction time	Baseline at diagnosis for deployable predictors; survival months explicitly classified as post-baseline.
Outcome	Recorded all-cause death with survival months; alive observations treated as right-censored.
Partitioning	Stratified 80:20 split before all fitted preprocessing in corrected analyses.
Partition manifest	Primary train-test assignment generated as a supporting CSV for exact reproduction.
Resampling	Legacy pre-split oversampling reproduced for audit; corrected oversampling restricted to training.
Preprocessing	Scaling and one-hot encoding fitted on training data only.
Hyperparameter tuning	Five-fold cross-validation within training partition; test set used once.
Uncertainty	1000 bootstrap test-set resamples for AUROC, balanced accuracy, and C-index; Wilson intervals for sensitivity and specificity.
Calibration	60-month predicted and Kaplan-Meier observed risks reported by quintile.
Code and environment	Executable script, pinned requirements, random seeds, and machine-readable outputs supplied.
Validation type	Explicitly described as internal random holdout; no external-validation claim.
Unsupported claims excluded	No TNBC, diagnostic, hospital, workflow, economic, or deployment claim.

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